

Module 1

1.1 Basics of AI, ML and DL

What is Artificial Intelligence (AI)?

Artificial Intelligence is the field of building machines or systems that can perform specific tasks that would normally require human intelligence, without needing a human to do them directly. It is essentially a system that has human-like intelligence without needing the physical body of a human. In simple words, AI is about giving machines the intelligence they need to carry out tasks on their own — such as reasoning, understanding language, recognizing images, or making decisions.

Example: Smart assistants like Siri or Alexa.

What is Machine Learning (ML)?

Machine Learning is a branch of Artificial Intelligence where machines are trained to perform tasks by learning from data patterns instead of being explicitly programmed with fixed rules. It learns the relationship between the data provided and understands the underlying pattern, so it can make predictions or decisions on new data. The more data it is trained on, the better it usually gets at recognizing patterns.

Example: Predicting stock market hikes based on historical price data.

What is Deep Learning (DL)?

Deep Learning is a subset of Machine Learning where a large amount of data is provided, and the system uses Neural Networks — models loosely inspired by the human brain — to understand complex patterns. Unlike traditional ML, DL automatically extracts the required features from raw data (such as images, audio, or text) without needing a human to define them manually. It usually requires large datasets and more computing power, but performs very well on complex, unstructured data.

Example: Face recognition, such as unlocking a phone using your face.

Relationship: AI is the broadest field. ML is a subset of AI. DL is a subset of ML. (AI → ML → DL, like nested circles — every DL system is ML, and every ML system is AI, but not the other way around.)

1.2 Difference Between AI, ML and DL

Feature	AI	ML	DL
Definition	Makes machines perform tasks which require human intelligence	Makes machines learn from data to perform tasks	Makes machines learn complex patterns using Neural Networks
Relationship	Broad field	Subset of AI	Subset of ML
Working	Uses different techniques to achieve intelligence	Learns patterns by itself when data is fed	Learns patterns through multiple neural-network layers
Data needed	Varies by technique used	Moderate, usually structured data	Large amounts of data, often unstructured
Example	Smart Assistant	Predicting stock market hikes	Face Recognition

1.3 Real World Applications

- **Recommendation systems** — Netflix, Spotify and YouTube use AI to suggest shows, music, and videos based on what you've watched or listened to before.
- **Voice assistants** — Siri, Alexa and Google Assistant understand spoken commands and respond using natural language processing.
- **Fraud detection** — Banks use ML models to spot unusual transaction patterns and flag possible fraud in real time.
- **Medical diagnosis** — AI helps doctors by reading X-rays and MRIs to detect diseases like cancer earlier and more accurately.
- **Self-driving cars** — Combine sensors and cameras (perception) with AI models (decision-making) to drive safely without human control.
- **Chatbots & customer support** — Automated conversational agents handle common customer questions, saving time for both companies and customers.
- **Translation** — Google Translate and similar tools use deep learning to translate text and speech between languages in real time.